

Using Virtual Reality to Promote Pro-environmental Consumption

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ABSTRACT

Virtual reality (VR) has emerged as an effective method for encouraging eco-friendly behaviors. The present study aimed to investigate the effectiveness and underlying mechanisms of VR in promoting pro-environmental consumption. The results indicated that, when compared to traditional mediums such as 2D video and printed material, VR can improve individuals' pro-environmental consumption. This favorable effect of VR was found to be mediated by the inclusion of nature in self (INS). Additionally, this study also demonstrated that individuals' green values act as the boundary condition.

Keywords: Virtual reality, pro-environmental consumption, inclusion of nature in self, green values, attitudes, buying.

Index Terms: J.4 [Social and Behavioral Sciences]; K.4.4 [Computers and Society]: Electronic Commerce

1 INTRODUCTION

In recent years, global environmental issues have become increasingly pressing, posing challenges to human survival and development. Many environmental problems originate from human consumption activities. Therefore, understanding how to make consumers care about nature and change their attitudes and behaviors toward the environment in their consumption has become a prominent research topic. With the advancements in digital technology, combining virtual reality (VR) technology with traditional strategies has become an important avenue for innovative green marketing [1]. Many well-known environmental organizations have already implemented VR in their consumer-oriented fundraising campaigns and have achieved better results, such as *World Wildlife, Charity: Water, and Animals Asia Foundation*. However, despite the increasing interest within the non-profit sector, academic research on the use of VR to enhance pro-environmental behavior is still in its early stages [2][3]. Therefore, this study aims to integrate VR with consumer applications and use experimental methods to explore VR's potential to improve pro-environmental consumption compared to traditional mediums such as 2D videos and printed materials [4].

This study has three objectives. The first purpose is to deepen our understanding of the impact of VR on pro-environmental consumption. Second, this study examines the internal mechanism of interaction between VR and pro-environmental consumption, that is, the mediating role of the inclusion of nature in self (INS). The self-expansion theory suggests that people spontaneously integrate new knowledge, perspectives, abilities, identities, and resources of others into their existing self-schemas and beliefs in order to enrich their self-concepts [5]. Nature can also become integrated into the self-concept and become a part of one's self through self-expansion, and this process is called the inclusion of nature in self [6]. Third, this study evaluates the moderating effect of an individual's green values on the relationship between VR and

pro-environmental consumption. We referred that VR would only affect the pro-environmental behavior of consumers with lower levels of green values.

The present study examines pro-environmental consumption at both the attitude and buying levels, as they are considered reliable predictors. We formulate the following hypothesis:

Hypothesis 1: VR nature experiences will enhance pro-environmental consumption attitudes (H1a) and promote pro-environmental buying (H1b) when compared to traditional mediums. **Hypothesis 2:** The INS will mediate the influence of VR vs. traditional mediums on both pro-environmental consumption attitudes (H2a) and pro-environmental buying (H2b), respectively. **Hypothesis 3:** Green values of an individual will moderate the effect of VR vs. traditional mediums on pro-environmental consumption.

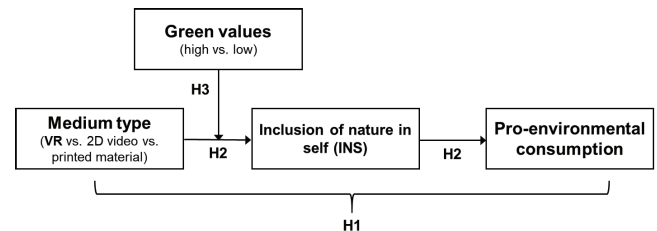


Figure 1. Conceptual framework of the present study.

2 METHODS

The present study adopted a single-factor (Medium type: VR vs. 2D video vs. printed material) between-subjects design. Based on the calculation of G*Power 3.1, a total of 175 participants were recruited and randomly assigned to one of the three conditions.

The experiment was conducted in a quiet laboratory. In the VR condition, participants wore Oculus Quest 2 VR headsets and experienced walking in a forest environment (using the Nature Treks application). Participants in the 2D video condition viewed a pre-recorded 2D video version of walking in the same forest on a computer screen, while participants in the printed material condition viewed one printed textual and graphic material about the forest ecosystem. Unlike the 2D video and printed material conditions, where information was accessed in the form of viewing, participants in the VR condition were immersed in a 3D, 360-degree virtual forest environment to receive information.

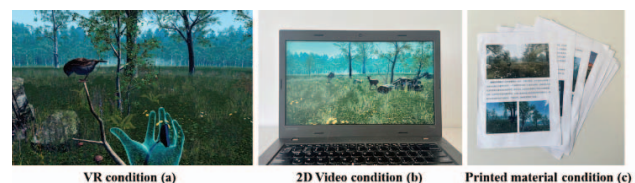


Figure 2. Examples of the VR (a), 2D video (b), and printed material (c) conditions.

We measured participants' green values before the formal experiment began (Cronbach's $\alpha = 0.87$). After participants received the stimuli, their perception of INS (Cronbach's $\alpha = 0.87$) and pro-environmental consumption attitudes (Cronbach's $\alpha =$

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0.88) were measured. Participants then engaged in a pro-environmental product buying task [7]. They were instructed to imagine that they were planning to purchase a notebook and had to choose between two notebook options. One option emphasized its eco-friendly features, while the other highlighted its performance-focused features. To encourage the real performance of the participants, a random sample was selected for the actual rewards based on their choices. We also measured participants' mood and awe and served them as control variables in the analysis.

3 RESULTS

3.1 The Main Effects

The ANCOVA was conducted and found that participants' attitudes were significantly higher in the VR condition compared to the 2D video condition ($F(1, 113) = 7.62, p = 0.009$) and the printed material condition ($F(1, 111) = 18.26, p < 0.001$). Therefore, H1a is supported. In terms of buying, a chi-square analysis yielded a significant difference in choice patterns across the three conditions ($\chi^2(2) = 8.97, p = 0.011$). As expected, the percentage of participants who chose an eco-friendly notebook to buy in the VR condition was higher than those either in the 2D video condition ($\chi^2(1) = 2.03, p = 0.043$) or in the printed material condition ($\chi^2(1) = 2.96, p = 0.003$). Thus, H1b is supported.

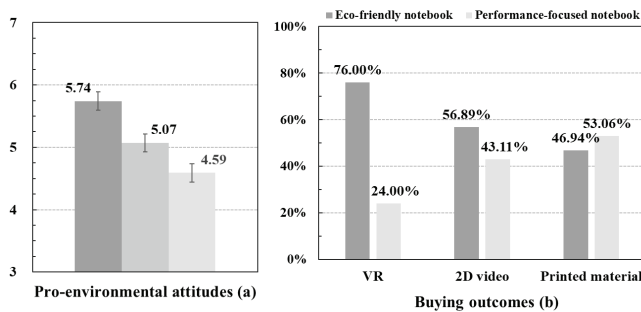


Figure 3. Comparisons of pro-environmental consumption attitudes (a), and buying outcomes (the percentage of choosing eco-friendly notebooks).

3.2 The Mediating Role of Inclusion of Nature in Self

The mediation analyses (using Mplus 8.3) were conducted. We dummy-coded medium type, using the VR condition as the reference group (dummy 1: VR = 0, 2D video = 1, printed material = 0; dummy 2: VR = 0, 2D video = 0, printed material = 1). The bias-corrected bootstrapping method with 5000 resamples was employed to estimate 95% confidence intervals. The obtained model results are presented in Figure 4.

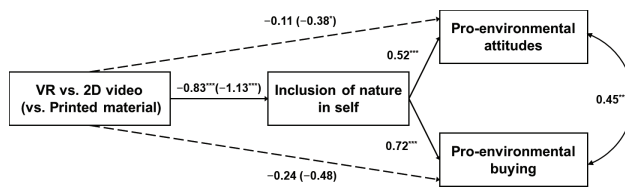


Figure 4. The mediation model of the present study. The values outside the parentheses represent the path coefficients comparing the VR to 2D video (dummy 1), while the values inside the parentheses represent the path coefficients comparing the VR to printed material (dummy 2).

Regarding pro-environmental consumption attitudes, VR (compared to 2D video and printed material) achieved significant indirect effects on attitudes through INS (dummy 1: $B = -0.83, CI_{95\%}$

$= [-1.25, -0.41]$; dummy 2: $B = -1.13, CI_{95\%} = [-1.56, -0.70]$). Similar findings were observed in analyses of pro-environmental buying. VR (compared to 2D video and printed material) had significant indirect effects on buying through INS (dummy 1: $B = -0.60, CI_{95\%} = [-1.07, -0.23]$; dummy 2: $B = -1.06, CI_{95\%} = [-1.62, -0.63]$). Therefore, H2a and H2b are supported simultaneously.

3.3 The Moderating Role of Green Values

We then incorporated green values into the original mediation model and conducted moderated mediation analyses (using Mplus 8.3). As expected, for attitudes, the conditional indirect effect of medium type through INS was significant for participants low in green values (-1 SD; dummy 1: $B = -0.59, CI_{95\%} = [-0.92, -0.33]$; dummy 2: $B = -0.82, CI_{95\%} = [-1.20, -0.52]$), but not for those high in green values ($+1$ SD; dummy 1: $B = -0.14, CI_{95\%} = [-0.38, 0.13]$; dummy 2: $B = -0.18, CI_{95\%} = [-0.42, 0.09]$). The same patterns also exist for buying. The indirect effect of medium type on buying through INS was significant for participants low in green values (-1 SD; dummy 1: $B = -0.84, CI_{95\%} = [-1.31, -0.46]$; dummy 2: $B = -1.19, CI_{95\%} = [-1.70, -0.73]$). In contrast, this mediation disappeared for participants high in green values ($+1$ SD; dummy 1: $B = -0.20, CI_{95\%} = [-0.56, 0.16]$; dummy 2: $B = -0.26, CI_{95\%} = [-0.67, 0.12]$). These findings suggested that the mediating process by which VR influences pro-environmental consumption through INS could be moderated by participants' green values. Therefore, the H3 is supported.

4 GENERAL DISCUSSION

The present study investigated the impact, underlying mechanisms, and boundary conditions of VR on pro-environmental consumption. The findings indicated that VR, when compared to traditional mediums such as 2D video and printed material, can effectively enhance consumers' pro-environmental attitudes and buying, and this effect is mediated by INS. Additionally, this study identified the individual trait of green values as a boundary condition, indicating that individuals with higher levels of green values are not influenced by the mediating process of VR through INS on their pro-environmental consumption. The findings of this study will enrich the existing literature [3][8] on VR and its applications in pro-environmental contexts by revealing whether, how, and when VR technology is effective for increasing pro-environmental consumption. This study can also inform social marketers of the benefits and limitations offered by VR technology and opens up multiple avenues for further application of VR and other emerging technologies in green marketing.

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